Statistics

Statistical Modeling and Learning (STM 2102, 4.5 credits)

The University of Austin

Instructor: David Puelz
Course Webpage: https://github.com/dpuelz/StatisticalModeling
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Meeting Schedule:

Section 1	M/W/F	11:30a–12:45p
Section 2	T/Th/F	10:00a–11:15a

Office Hours: Please consult the webpage

Course Description

This course trains students to wield data as both a scientific and predictive tool with an applied focus on business and economics. It develops mastery in two complementary approaches: supervised machine learning, which builds powerful models for prediction and decision making, and causal inference, which uncovers credible cause and effect relationships. Students will learn to design empirical studies, implement algorithms, and critically evaluate when data reveals insight.

Course Objectives

- Understand the fundamentals of statistical modeling and supervised learning
- Fit and interpret regression models for prediction and inference
- Apply statistical methods to real-world data using appropriate software
- Evaluate model performance and understand the bias-variance tradeoff

Required Readings

The first book is available for free online. The second book is supplemental reading for the causal inference parts of class.

- *Introduction to Statistical Learning* (ISL) — Gareth James, et al.
- *Mastering 'Metrics* (MM) — Joshua D. Angrist and Jorn-Steffen Pischke

Course Cadence

There will be 5 quizzes and 5 homework assignments. The quizzes will be on the Fridays of weeks 2, 4, 6, 8, and 10. The homeworks will be due at the start of class on the same Fridays as the quizzes—11:30a for Section 1 and 10:00a for Section 2. The quiz content will be related to the homework, and we will mark up the quizzes in class directly after finishing the quiz. We will have a final exam during the scheduled exam time (on week 11 of the course).

Evaluation

The class grade is comprised of the following:

- Homework (20%)
- Exam (30%)
- Quizzes (50%)

Topics

This is a course on supervised learning. We will cover the following topics: (*i*) introduction and bias-variance tradeoff, (*ii*) inference for regression, (*iii*) multiple regression, (*iv*) categorical predictors and interactions, (*v*) assumptions and diagnostics, (*vi*) nonlinear regression, (*vii*) time series regression, (*viii*) logistic regression, (*ix*) model selection and penalized regression, (*x*) trees, ensembles, and neural networks, and (*xi*) causal inference.

Below is a timeline for the semester. It is subject to change as we make our way through these topics. Please also pay attention to the course website. I will include supplemental readings beyond those in our textbooks.

Week	Topics	Reading
1 (Aug 31)	Intro and bias-variance tradeoff	ISL: Ch 1, 2
2 (Sep 7)	Inference for regression + Multiple regression	ISL: Ch 3.1–3.2, MM: Ch 2
3 (Sep 14)	Categorical predictors and interactions	ISL: Ch 3.3
4 (Sep 21)	Assumptions, diagnostics + Nonlinear regression	ISL: Ch 3.3, 7.1
5 (Sep 28)	Time series regression	ISL: Ch 3 (supplemental)
6 (Oct 5)	Logistic regression	ISL: Ch 4.1–4.3
7 (Oct 12)	Model selection and penalized regression	ISL: Ch 6.1–6.5
8 (Oct 19)	Trees, ensembles, and neural networks	ISL: Ch 8.1–8.3, 11.1–11.3
9 (Oct 26)	Causal inference	MM: Ch 1, ISL: Ch 3.2
10 (Nov 2)	Causal inference	MM: Ch 3–5
11 (Nov 9)	*final exams Nov 10–17*	

The term runs Monday, August 31 through Monday, November 9, with final exams November 10–17. Labor Day (Monday, September 7) and Veterans Day (Wednesday, November 11) are university holidays. Add/drop closes Thursday, September 10; the last day to withdraw is Wednesday, October 14.

Accessibility Statement

Please review the University Accessibility Statement in the academic catalog. Qualified students may request reasonable accommodations by contacting the Provost's Office at accommodations@uaustin.org. Disclosure is voluntary, but accommodations are not provided without formal disclosure. The University will make reasonable accommodations for students with disabilities in compliance with Section 504 of the Rehabilitation Act and the Americans with Disabilities Act. The purpose of accommodations is to provide equal access to educational opportunities for eligible students with academic and/or physical disabilities.

Attendance Policy

Attendance is mandatory. If you don't show up, you will fail the course. Each student may miss up to 10% of classes for any reason—no excuse required and without penalty—corresponding to 1, 2, or 3 classes in a 1.5, 3.0, or 4.5 credit course, respectively. *This course is 4.5 credits, so three absences are allowed without penalty.* Any additional absences will result in a final grade penalty of 12% (1.5 credits), 6% (3.0 credits), or 4% (4.5 credits), regardless of the reason. Missing more than 25% of classes—including the allowable absences—without an approved medical excuse will result in failing the course. Being more than 20 minutes late to class counts as an unapproved absence.

Absences are excused only when approved under university policy: obligations of university interest (Provost); medical or bereavement emergencies, or civic obligations such as jury duty or military service (Students & Community). Submit a request with supporting documentation within three business days of the absence. Excused absences permit makeup of missed coursework without penalty; unexcused absences do not permit makeup of missed in-class work.

Showing Work

On all course deliverables, there is an expectation of showing how you got to an answer. If you show a partially correct sequence of steps to a wrong answer, you may be awarded partial credit (at the discretion of the instructor). If you only write down the correct answer, you will not receive any credit. I will not award credit to random guessing and shoddy, vague, or unclear supporting work. This is part of the pedagogical design—demonstrate to me that you know what you're doing!

Late Policy

Sometimes we have bad days, bad weeks, and bad terms. In an effort to accommodate any unexpected, unfortunate personal crisis, I have built a grace policy into the course: that is, a one-time, three-day grace period for one homework assignment. You do not have to utilize this policy, but if you find yourself struggling with unexpected personal events, I encourage you to e-mail me as soon as possible to notify us that you are using the grace policy. *All other late assignments will be penalized 20 percentage points per day or partial day that they are late.*

STEM Center Learning Objectives

This is an intellectual foundations course that contributes to the following STEM Center learning objectives:

→ Speak, read, and write in mathematical language.

→ Formulate and solve quantitative problems using appropriate technical language, models, and techniques.

→ Collect and analyze data using appropriate technology tools.

→ Communicate technical solutions clearly through writing, speaking, and visualization.

Final Course Notes: AI and Academic Misconduct

You are encouraged to use AI tools to the fullest extent on homework and coding—as collaborators, tutors, and calculators. Learn as much as you can with them. Quizzes and the exam are how I assess what you have actually learned; treat them accordingly. I have zero tolerance for academic misconduct on quizzes or the exam (including copying from others or unauthorized materials). A violation earns an automatic zero on that deliverable.